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Time left 1:29:02

Question 1 Not yet answered Marked out of 1.00 Flag question

Let A and B be 3×3 matrices with $\det A = 2$ and $\det(2B^T) = 24$. Then $\det(-2A^{-1}B^{-1}A^T) =$

Select one:

- A. -1/6
- B. -2/3
- C. unknown, because there is not enough information to find the value
- D. 6/3

Question 2 Not yet answered Marked out of 1.00 Flag question

Let $f(x) = \begin{cases} \cos(x^{-2}) & , \quad x \neq 0 \\ 2 & , \quad x = 0 \end{cases}$ and let $g(x) = 2^{f(x)} \cdot \sin^2 x$

Which of the following must be true?

Select one:

- A. g is discontinuous at 0 although 0 is in the domain of g and $\lim_{x \to 0} g(x)$ exists
- B. g is discontinuous at 0 since 0 is not in the domain of g
- C. g is discontinuous at 0 since $\lim_{x \to 0} g(x)$ does not exist
- D. g is continuous at 0

Question 3 Not yet answered Marked out of 1.00 Flag question

A total of x feet of fencing is to form three sides of a level rectangular yard. What is the maximum possible area of the yard, in terms of x?

Select one:

- A. x^2
- B. $2x^2$
- C. $\frac{x^2}{8}$
- D. $\frac{x^2}{4}$
- E. $\frac{x^2}{2}$

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Question 4 Not yet answered Marked out of 1.00 Flag question

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Let $f(x) = \sqrt{|x+4|-1}$, $g(x) = x^2$, $h(x) = x - 3$. What is the expression for $(f \circ g \circ h)(x)$?

Select one:

- A. $\sqrt{x^2 - 6x + 12}$
- B. $\sqrt{|x^2+4|+1} - 3$
- C. $(\sqrt{|x+4|-1})(x^2)(x-3)$
- D. x with $x \geq 0$
- E. $|x+4| - 4$

Question 5 Not yet answered Marked out of 1.00 Flag question

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a linear transformation such that $T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 4 \\ 5 \end{bmatrix}$ if $T(\mathbf{e}_1) = 3\mathbf{e}_1$, find $T(\mathbf{e}_2)$. Here $\{\mathbf{e}_1, \mathbf{e}_2\}$ is the standard basis of $\mathbb{R}^2$.

Select one:

- A. $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$
- B. $\begin{bmatrix} -2/3 \\ -5/3 \end{bmatrix}$
- C. $\begin{bmatrix} 2/3 \\ 5/3 \end{bmatrix}$
- D. $\begin{bmatrix} -2/3 \\ 5/3 \end{bmatrix}$

Question 6 Not yet answered Marked out of 1.00 Flag question

The variance of a constant c is

Select one:

- A. 0
- B. undefined
- C. 1
- D. c

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Time left 1:28:02

Suppose $f : [0, \infty) \to \mathbb{R}$, find $f(4)$ if

$$\int_0^{x^2} f(t)\,dt = x\cos \pi x .$$

Select one:

- A. $\sqrt{12}$
- B. $\sqrt[3]{12}$
- C. $\frac{1}{2}$
- D. None of the other choices is correct
- E. $\frac{1}{4}$

Question 8 Not yet answered Marked out of 1.00 Flag question

Let x and y be real numbers such that $\begin{bmatrix} 8 \\ 8 \end{bmatrix} = x\begin{bmatrix} 6 \\ -9 \end{bmatrix} + y\begin{bmatrix} 3 \\ 3 \end{bmatrix}$. Find the value of x.

Select one:

- A. 1
- B. 0
- C. -2
- D. 2

Question 9 Not yet answered Marked out of 1.00 Flag question

A university's mathematics department has 5 professors and will offer 15 different courses next semester. Each professor will be assigned to teach exactly 3 of the courses, and each course will have exactly one professor assigned to teach it. If any professor can be assigned to teach any course, how many different complete assignments of the 5 professors to the 15 courses are possible?

Select one:

- A. $\frac{15!}{3!^5}$
- B. $\frac{15!}{3!}$
- C. $15^5/3!$
- D. $15^5 - 3^{15}$
- E. $\frac{15!}{3!}$

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Question 10 Not yet answered Marked out of 1.00 Flag question

Time left 1:27:36

Let $X_1, X_2, \ldots, X_n$ be n independent random variables sharing the same probability distribution with mean μ and variance σ^2 (> 1). Then, as n tends to infinity, the distribution of the standardized random variable

$$\frac{X_1 + X_2 + \cdots + X_n - n\mu}{\sqrt{n}\sigma}$$

converges to

Select one:

- A. a normal distribution with parameters 0 and 1
- B. a normal distribution with parameters μ and σ^2
- C. a Poisson distribution
- D. an exponential distribution

Question 11 Not yet answered Marked out of 1.00 Flag question

Let X be a normal random variable with parameters 3 and 4. Then, which of the following random variables is a standard normal random variable?

Select one:

- A. $(X+2)/3$
- B. $(X+3)/2$
- C. $(X-2)/3$
- D. $(-X+3)/2$

Question 12 Not yet answered Marked out of 1.00 Flag question

Let A be a square matrix and E be an elementary matrix of the same size as A. Which of the following is/are true?

I. If A is invertible, then EA is invertible.
II. If A has an LU decomposition, then EA has an LU decomposition.
III. If A has a zero row, then EA has a zero row.
IV. If A has a zero column, then EA has a zero column.

Select one:

- A. II, III and IV only
- B. I and IV only
- C. I and II only
- D. III only

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Consider a continuous random variable X whose probability density function takes the following form:

$$p(x) = \begin{cases} Kx^2 & \text{, if } 0 \le x \le 1, \\ 0 & \text{, if otherwise,} \end{cases}$$

where K is an unknown constant. Then, $P(-1/2 \le X \le 1/2)$

Time left 1:27:19

Select one:

- A. cannot be determined
- B. is equal to $1/4$
- C. is equal to $1/2$
- D. is equal to $1/8$

Question 14 Not yet answered Marked out of 1.00 Flag question

Compute $\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{|x|+|y|}$

Select one:

- A. The limit does not exist
- B. 2
- C. 0
- D. 1

Question 15 Not yet answered Marked out of 1.00 Flag question

Which of the following matrices is in reduced row echelon form?

Select one:

- A. $\begin{bmatrix} 1 & 3 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$
- B. $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 0 & 1 \end{bmatrix}$
- C. $\begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 1 \end{bmatrix}$
- D. None of the others
- E. $\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}$

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Question 16 Not yet answered Marked out of 1.00 Flag question

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Find the area of the surface of revolution obtained by rotating the given parametric curve about the y-axis:

$x = \cos(t), \quad y = \sin(t), \quad 0 \le t \le \pi/2.$

Select one:

- A. 4π square units
- B. 6π square units
- C. 8π square units
- D. 2π square units
- E. 16π square units

Question 17 Not yet answered Marked out of 1.00 Flag question

Consider two urns, the first of which contains 3 black balls and 2 white balls, and the second of which contains 2 black balls and 3 white balls. Randomly pick an urn and then randomly pick a ball from the chosen urn. Given that the chosen ball is black, what is the probability that it is chosen from the first urn?

Select one:

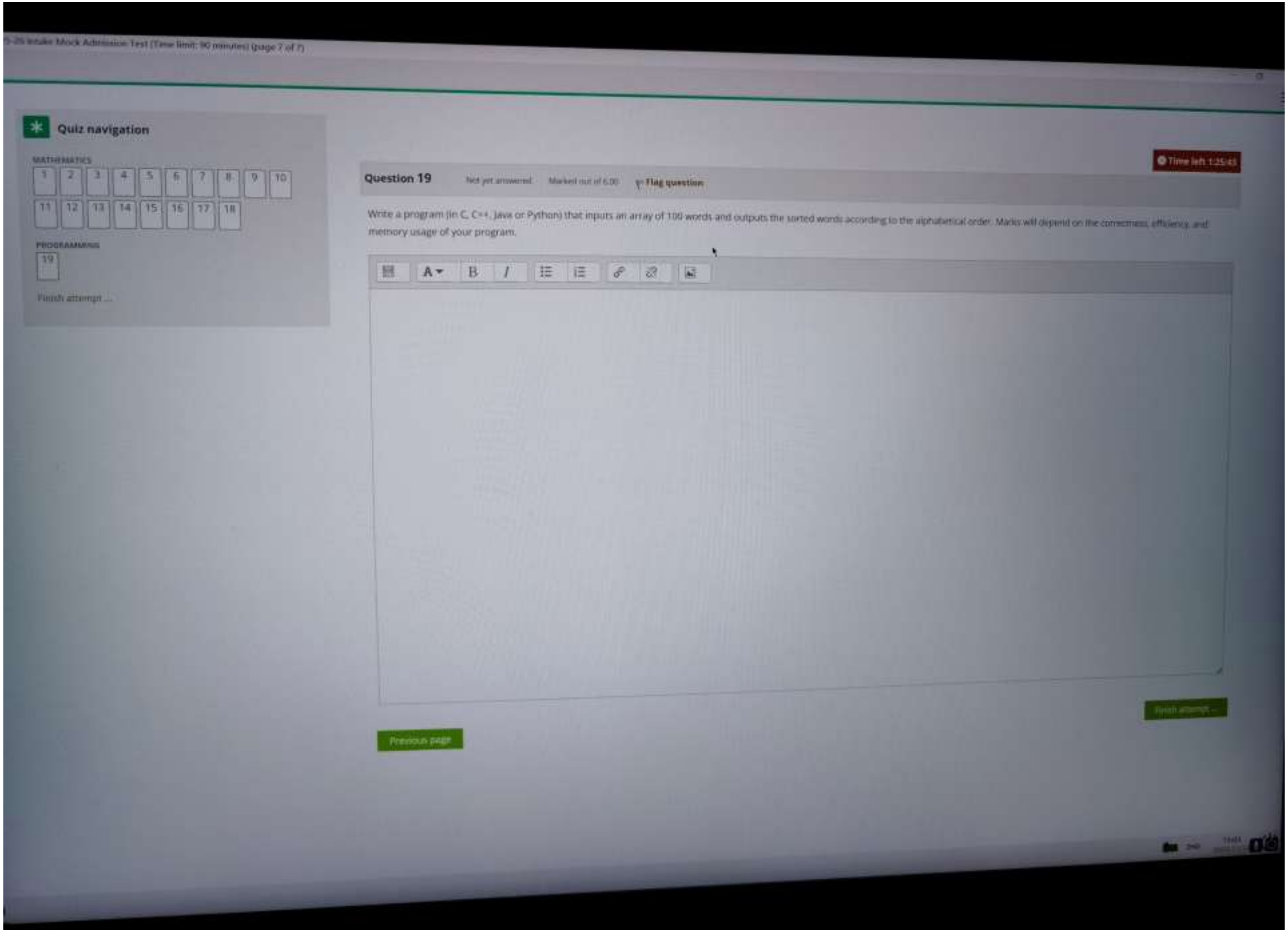
- A. $2/5$
- B. $1/2$
- C. $2/3$
- D. $3/5$

Question 18 Not yet answered Marked out of 1.00 Flag question

Let A be a 5×5 matrix such that the sum of the entries of each row is equal to 5. Select a number that MUST be an eigenvalue of A.

Select one:

- A. 4
- B. 5
- C. 1
- D. 2
- E. 0


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Question 19
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Marked out of 6.00
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Write a program (in C, C++, Java or Python) that inputs an array of 100 words and outputs the sorted words according to the alphabetical order. Marks will depend on the correctness, efficiency, and memory usage of your program.
Finish attempt ...
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